

RUPALI MEHTA

+91 - 9975852367 | mehtarupali78@gmail.com | <https://github.com/mehtarupali78> | <https://www.linkedin.com/in/rupali-mehta-003/> | <https://medium.com/@mehtarupali78>

SUMMARY

Senior Full-Stack Engineer with over **8 years of experience** architecting, scaling, and delivering high-performance enterprise platforms across **FinTech, HealthCare, Media/OTT, and EdTech** domains. Expert in engineering resilient **Node.js microservices**, optimizing complex cloud infrastructure, and building fluid user interfaces leveraging the **JavaScript, React/Next.js** ecosystem and **AWS cloud architecture**. Proven track record of building scalable web applications, improving application performance, and delivering real-time platforms supporting **100K+ concurrent users**.

SKILLS

Core Languages & Runtimes: NodeJS, NestJS, ExpressJS, REST API Development, Serverless framework, GraphQL, Python, WebSockets, JWT Authentication, Authentication & Authorization, API Integration, Puppeteer

Frontend Architecture: ReactJS, NextJS, TypeScript, JavaScript (ES6+), Redux, Zustand, Angular, Context API, HTML5, CSS3, Tailwind CSS, Material UI, AG-Grid, Responsive Design, Cross-Browser Compatibility, Component Reusability, Frontend Optimization, Code Splitting, Lazy Loading

Testing & Quality: Jest, React Testing Library, Enzyme, Unit Testing

Cloud & Infrastructure: AWS (Lambda, EC2, S3, CloudFront, API Gateway, Cognito, AppSync, RDS, SQS, SNS), CI/CD Pipelines, Git/GitHub, GitHub Actions, Serverless Framework, Jira, Twilio, OpenVidu, Brightcove

Databases & Data Layers: PostgreSQL, DynamoDB, Redis, RabbitMQ, Prisma ORM

System Design & Methodologies: Distributed Systems, Event-Driven Architecture, Scalable Architecture, Microservices, Asynchronous processing, System Design, Message Queues, Micro-Frontend Architecture, Edge Caching, High Availability, Performance Tuning, Real-Time Applications, Agile/Scrum

AI & Emerging Technologies: LLM Applications, Retrieval-Augmented Generation (RAG), AI Agent Workflows, Prompt Engineering, AI-assisted code generation & Optimization

EXPERIENCE

The Transformation Group | Senior Software Developer.

April 2025 – Present

Project: Impact Studio (Five-Year Plan Module — Employer Premium Contribution Reporting)

- Architected the frontend ecosystem using **React** for the **Five-Year Plan module** within **Impact Studio**, accelerating automated plan forecasting and data-processing throughput by **12%**.
- Designed and engineered automated reporting tools utilizing the **Domo data layer** to analyze historical corporate data and generate financial projections on employer premium contribution limits.
- Built and deployed a localized **custom AI Agent automation** workflow using **Node.js** and the **OpenAI SDK**; automated the extraction and compilation of complex multi-factor financial requirements from Excel sheets directly into deterministic, typed **TypeScript** calculation modules.
- Orchestrated automated self-correction verification loops with **Jest** to cross-validate generated code outputs against legacy Excel math configurations, slashing manual engineering translation time by **30%**.
- Structured deterministic client-side state management layers with Zustand, improving predictability metrics by **22%** while smoothly mapping factor-driven AI calculations into high-performance data-visualization graphs.
- Spearheaded **rendering optimization** and asset bundle profiling across intensive data-visualization views, dropping structural code complexity by **18%**.

Macrometa | Technical Staff

August 2024 – February 2025

Project: PhotonIQ Prerender Engine

- Developed **PhotonIQ Prerender**, a scalable edge-rendered caching solution for single-page applications, improving **Core Web Vitals** and **SEO performance** by **25%**.
- Optimized legacy backend services and platform performance, reducing overall **system latency by 30%** and lowering cloud infrastructure costs by **15%**.
- Built scalable asynchronous **event-driven systems** to improve global data availability and support distributed application scalability across multiple regions.

Deutsche Bank | Senior Software Developer

August 2023 – August 2024

Project: LDCUI Banking Dashboard & Client Onboarding Platform

- Developed real-time banking dashboards using **ReactJS**, **AG-Grid**, and **TypeScript** to safely display financial transaction data across varied clients.

- Integrated high-performance frontend interfaces directly with enterprise backend layers structured on **Java** and the internal **VAS framework**.
- Migrated legacy banking modules into a decoupled frontend architecture, improving **feature delivery speed by 25%** and reducing client-side **UI latency by 20%**.
- Automated client verification components and customized **lazy loading modules** to improve overall application maintainability.

Particle41 LLP | Software Developer
August 2019 – August 2023

Project: Forte Lessons (LMS) & OTT Infrastructure

- Architected scalable **OTT** and **live-streaming web applications** using **ReactJS**, **NextJS**, **GraphQL**, **brightcove** and **WebSockets**, sustaining **100K+ concurrent users** during large-scale digital music festivals.
- Engineered **Forte Lessons** virtual learning system using **WebRTC**, **Serverless Framework**, **twilio** and **AppSync**, driving a **20% increase** in student retention.

Project: Computer Share (Automate Boat)

- Built an automation tool from scratch utilizing **AWS Lambda**, **NestJS**, **Serverless Framework** and **OCR** text parsing to process tax returns automatically across multiple state portals, cutting manual workflow times by **40%**.

Project: Classroom Door & Circle Media

- Developed a multi-user virtual classroom frontend with an interactive **Fabric.js whiteboard layer**, **Firebase Realtime DB**, and **OpenVidu video calling** integrations.
- Managed automated **CI/CD pipeline** deployments using **bitbucket** and **AWS infrastructure automation**, maintaining **99.9% environment availability**.

Terse Soft PVT LTD | Software Developer
August 2018 – August 2019

Project: Aurovine, Credentia & Multi-Tenant Infrastructure

- Designed distributed media streaming pipelines utilizing decentralized topologies, **React.js** frontend components, and **ExpressJS / PostgreSQL** architectures, reducing transaction packet drop frequencies by **12%**.
- Developed immutable identity authentication layers leveraging **IPFS** protocol configurations with **ExpressJS** routers and **PostgreSQL** data stores, mitigating system interaction fraud metrics by **20%**.
- Engineered modular multi-tenant e-commerce architectures with scalable **React.js** layouts and secure **ExpressJS** API backends, producing a **15% expansion** in net operational transaction capacity.
- Built a custom **B2B** manufacturing **CRM** system using **React.js**, **ExpressJS**, and **PostgreSQL** to automate product consignment tracking and record complex inventory states in real time.

EDUCATION & CERTIFICATIONS

- B.E. in Computer Science & Engineering | Siddhivinayak Technical Campus, Khamgaon (2014 – 2017)
- Diploma in Computer Science & Engineering | Siddhivinayak Technical Campus, Khamgaon (2011 – 2014)
- AWS Certified Cloud Practitioner | Amazon Web Services (Validation Period: 2023-2026)